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Editors
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Dear Editors,

Please consider our manuscript, "Confidence-Shielded, Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection," for publication in *Infrared Physics & Technology*.

Infrared–visible fusion can expose thermal evidence to object detectors, but the same fused image does not benefit every detector. We formulate input selection as a label-free arbitration problem. The proposed Confidence-Shielded, Evaluator-Conditioned Fusion (CS-ECF) evaluates visible, consensus-detail fusion (CDF), PIAFusion, and SPGFusion candidates with the target detector. A fused input is used only when its top-five confidence reaches 1.5 times that of visible; otherwise, visible is retained as a safety fallback.

Across 12 confirmatory checkpoints on LLVIP, M3FD, and an official-test subset of DroneVehicle, the prespecified dataset-balanced AP50 gain was 0.0133, with a training-run-by-group hierarchical 95% interval of [0.0026, 0.0261]. Removing the shield increased the magnitude of the worst observed AP50 loss from 0.0040 to 0.0584. Nine of 12 checkpoint point estimates were positive, but no individual contrast survived Holm correction and the sign test was not significant ($p = 0.073$).

Against the strongest fixed baseline, Always CDF, the dataset-balanced difference was 0.0102 AP50, with a training-run-by-group interval of [-0.00057, 0.02980]. The interval included zero; the point estimate was concentrated in M3FD and DroneVehicle and was not present on LLVIP. We therefore describe a conditional risk-control contribution, not dominance over the best fixed fusion input.

Three additional evaluators were not used in policy construction. Two COCO-pretrained models improved AP50 over visible by 0.0213 and 0.0250, with scene-group intervals above zero, but neither separated from Always CDF. An LLVIP-fine-tuned YOLOv9t then exposed a negative portability boundary: CS-ECF lost 0.0108 AP50 relative to visible, although the shield improved the no-shield selector by 0.0153. Its calibration error was about five times larger than those of the zero-shot evaluators, and 41.4% of its switched images degraded. We report this counterexample because confidence arbitration is conditional rather than universally portable.

This work addresses how thermal information can be admitted into a deployed detection system and reports the benefit, cost, and failure modes on real infrared–visible data. The M3FD gain is concentrated in the people class, LLVIP intervals cross zero, and the fine-tuned evaluator reverses the visible-relative sign. We therefore present CS-ECF as a bounded, evaluator-conditioned system, not as evidence of uniform improvement or universal transfer.

The manuscript is original, has not been published previously, and is not under consideration by another journal. An earlier, related manuscript by the same authors on evaluator-dependent fidelity-index rankings was previously submitted to this journal and was not accepted. It is not under consideration elsewhere. The present manuscript is a distinct method study; it does not reproduce the earlier fidelity-ranking claims, and all shared datasets and evaluator infrastructure are identified in the manuscript. All authors have approved the submission and declare no competing interests. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The manuscript includes an AI-use declaration and statements for data, code, CRediT contributions, and competing interests.

Thank you for your consideration.

Sincerely,

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